INCH-POUND

MIL-DTL-85097/1B(OS)
27 July 1998
SUPERSEDING
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31 October 1985

DETAIL SPECIFICATION SHEET

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS)
ROCKET CATAPULT MARK 16 MOD 1 (LOADED ASSEMBLY)

This specification sheet is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the Mark 16 Mod 1 Rocket Catapult, Loaded Assembly described herein shall consist of this specification sheet and the issue of MIL-DTL-85097 listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation.

1. SCOPE

1.1 Scope. This specification sheet covers the requirements unique to the Mk 16 Mod 1 Rocket Catapult, Loaded Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Other Government documents, drawings, and publications. The following drawing forms a part of this document to the extent specified herein. Unless otherwise specified, the issue is that cited in the solicitation (see 6.2).

SPECIFICATIONS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

AS 3550 Propellant, N-43 (U)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Division (Code 840), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377
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AS 3554 Propellant, N-44(U)

(Application for copies should be addressed to Commander, Naval Air Systems Command, Code 51122E, Washington, DC 20361.)

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

707AS129	Container Assembly, for Rocket Catapults
707AS130	Palletization Assembly, Rocket Catapults
707AS300	Assembly, Rocket Catapult, MK 16 Mod 1
707AS190	Test Chamber
707AS191	Test Setup Assembly
707AS192	Adapter

(Application for copies should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Code 8410P, 101 Strauss Avenue, Indian Head, MD 20640-5035.)

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND

OP 2165 Navy Transportation Safety Handbook for Hazardous Materials

(Application for copies should be addressed to the Commanding Officer, Naval Publications and Forms Center, Attn: 1053, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements in this specification sheet, unless in direct conflict with the basic specification, shall be considered additional, not substitute. The loaded rocket catapult shall conform to the documents specified herein and to Drawing 736AS300.

3.2 Preproduction.

3.2.1 Propellant aging. When tested in accordance with MIL-DTL-85097, the cartridge and sustainer propellants shall meet the requirements of table I or table II, as applicable.

3.2.2 Load. The rocket catapult shall withstand a static compression load of 18,000 pounds and a tension load of 8,700 pounds applied along the longitudinal axis at the attachment points without failure.

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TABLE I. *Propellant N-43 (AS 3550).*

Property	Temp (°F)	Zero time		7-Year Equivalent			
				0% Rel. humidity		22% Rel. humidity	
		Min	Max	Min	Max	Min	Max
S_m (maximum stress) psi	77	157	256	157	269	157	276
E_m (elongation at S_m) %	77	18	28	9	28	16	28
Y_m (Young's modulus) psi	77	1014	1946	1014	3094	1014	3114
B_r (strand burn rate at 1000 psi) in/sec	77	0.60	0.67	0.60	0.71	0.60	0.71

TABLE II. *Propellant N-44 (AS 3554).*

Property	Temp (°F)	Zero time		7-Year Equivalent			
				0% Rel. humidity		22% Rel. humidity	
		Min	Max	Min	Max	Min	Max
S_m (maximum stress) psi	77	116	193	116	202	116	208
E_m (elongation at S_m) %	77	30	50	14	50	26	50
Y_m (Young's modulus) psi	77	589	1093	589	1737	589	1745
B_r (strand burn rate at 1000 psi) in/sec	77	0.55	0.67	0.60	0.71	0.60	0.71

3.3 First article/production.

3.3.1 Propellant. The propellant shall conform to the requirements of AS 3550 and AS 3554 for the following parameters at -65, 70, and 160 °F: heat of explosion, burning rate, and physical properties (see 6.2 and 6.3).

3.3.2 Ballistic performance. The rocket catapult shall meet the requirements of table III when test-fired using an ejected weight of 315 pounds at 160 °F and 410 pounds at -40 and 70 °F.

3.3.2.1 Output cartridge pressure. The head end output cartridge shall produce pressure values as specified in table III, when fired in a test chamber conforming to Drawing 736AS190, using test setup assembly and adapter in accordance with Drawings 736AS191 and 736AS192, respectively. The head end output cartridge shall reach maximum pressure within 10 milliseconds of rocket motor ignition as defined in MIL-DTL-85097.

3.3.2.2 Acceleration. During the first 0.015 second after initiation, any initial pulse shall not exceed 19 g's or exceed 5 g's for more than 0.010 second.

3.3.3 Rocket motor marking. The rocket motor serial number shall be stamped in black ink on the outer diameter of the rocket motor housing.

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TABLE III. *Ballistic performance requirements.*

Parameter	-40 ± 5 °F		70 ± 5 °F		160 ± 5 °F	
	Min	Max	Min	Max	Min	Max
Catapult ignition delay (ms)	0	5	0	5	0	5
Onset (g/s)	-	250[1]	-	250[1]	-	350[1]
Acceleration (g)	-	19[1]	-	19[1]	-	20[1]
Separation velocity [2] (ft/s)	41	-	-	-	-	-
Rocket ignition time (ms)	0	25	0	25	0	20
Rocket burn time (ms)	235	385[1]	200	285	175	270
Rocket action time (ms)	290	425[1]	240	355	210	320
Rocket motor resultant thrust (max)	3,165	4,935	3,700	5,800	4,750	7,300[1]
Rocket motor impulse (lb-s)	1,027[1]	-	1,027[1]	-	1,027[1]	-
Output cartridge (psi max)	1,080	2,100	-	-	1,080	2,100

[1]System requirements.
[2]At end of catapult stoke measured over the interval from 31 to 43 inches travel.

4. QUALITY ASSURANCE PROVISIONS

4.1 Examination and tests.

4.1.1 Preproduction.

4.1.1.1 Environmental.

4.1.1.1.1 Vibration. For the AEPS covered by this specification sheet, the following test procedures shall be used in lieu of those specified in the general specification. Two rocket catapults shall be subjected to vibration testing. One rocket catapult shall be temperature conditioned for a minimum of 8 hours at −40 ± 5 °F and the other at 160 ± 5 °F. The rocket catapults shall be maintained at specified conditioning temperatures during these tests. If the rocket catapult must be removed from the conditioning chamber or if the rocket catapult temperature falls outside the specified temperature limits for more than 5 minutes, the testing shall stop and the rocket catapult shall be reconditioned. A reconditioning time of 2 minutes for every minute the rocket catapult spent outside the specified temperature envelope shall be required, up to a maximum of 8 hours. A continuous temperature history of the rocket catapult shall be kept for all conditioning and test time. Excitation shall be applied through the aircraft seat attachment fittings unless otherwise noted in the contract. Accelerometers to measure acceleration along the axis perpendicular to the longitudinal axis of the rocket catapult shall be placed midway between the attachment points. Longitudinal vibration shall be measured at either end of the rocket catapult.

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4.1.1.1.1.1 Resonance survey. Resonance surveys shall be conducted on both rocket catapults in three axes at the maximum and minimum conditioning temperatures. Input vibratory acceleration of 5 g's shall be applied successively along each of the axes noted above when sweeping between 10 an 499 hertz. Input vibratory acceleration of 3 g's shall be applied for sweeps between 500 and 2,000 hertz. All resonances shall be as noted. During the resonance survey a permanent, continuous record of input vibratory acceleration and of unit response to the input vibratory acceleration shall be kept. This record shall be made on an instrument acceptable to the Government and shall include the response along each of the three axes. Frequency shall be varied at a rate not greater than 0.25 octave per minute.

a. A resonance shall be defined as a frequency at which a clearly defined peak in the amplitude ratio (ratio of the output acceleration in any of the three igniter cartridge axes to the input acceleration as measured at the excitation attachment points) exceeds a value of 1.5 to 1.

b. If the resonance survey indicates a frequency band wherein the amplitude ratio exceeds 1.5 to 1 (as contrasted to a clearly defined peak) this entire band width shall be considered one resonant point. The minimum frequency band width which shall be considered a resonant band shall be 0.33 octave. If, also within this band width, there are clearly defined amplitude peaks, the most severe of these shall be considered an additional resonant point.

4.1.1.1.1.2 Endurance vibration. One rocket catapult shall be conditioned to -40 ± 5 °F and one to 160 ± 5 °F. Each rocket catapult shall be vibrated for 24 hours, 8 hours in each of the three axes. The vibration time in each axis shall be equally divided between the resonant frequencies noted in that axis during the resonant survey. A peak input acceleration of 5 g's shall be maintained at frequencies between 10 and 499 hertz and 3 g's for frequencies between 500 and 2,000 hertz. If no resonant frequency is detected in an axis, the rocket catapult shall be vibrated in that axis for a minimum of 2 hours at 5 g's input through the 10 to 499 hertz range and 6 hours at 3 g's input through the 500 to 2,000 hertz range. Total vibration time for the rocket catapults shall be 24 hours which shall include resonant and nonresonant frequencies, as applicable. At the conclusion of the tests, the rocket catapults shall be visually and radiographically inspected for damage without disassembly of the rocket catapult. The rocket catapults shall be fired at the same temperatures they were conditioned and vibrated.

4.2 Preproduction/production.

4.2.1 Conditioning temperatures. The conditioning temperatures and tolerances shall be as follows:

a. Cold: -40 ± 5 °F
b. Hot: 160 ± 5 °F

4.2.2 Conditioning period. Time for conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: not less than 6 hours.
b. Temperature cycling: not less than 6 hours.

5. PACKAGING

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5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Mk 16 Mod 1 Rocket Catapult is used to eject the pilot from the TA-4J, S-3B, and ES-3A aircraft. As the A-4E/L aircraft is a military aircraft, there is no commercial application for the MK 16 Mod 1 Rocket Catapult.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).

c. That the contract packaging requirements include the following: that rocket catapults be packed as specified on Drawings 707AS129 and 707AS130 , and that shipping containers be marked “Rocket Motors, Class B - Keep Fire Away” in accordance with NAVSEA OP 2165.

6.3 Propellant description. The following propellant information should be submitted with each production lot. The materials and components of each propellant batch should be listed with the manufacturer, lot number, and date of manufacture. The burning rates at 500, 1,000, and 1,500 psi performed at –65, 70 and 160 °F for each pressure (three samples at each temperature and pressure per bucket) should also be included. In addition, include the results of tensile strength and elongation tests at –65, 70 and 160 °F with a pull rate of 2 inches/minute (three samples at each temperature per bucket), heat of explosion (3 samples per bucket), and verification that ingredients were properly weighed and added to each mix (mix sheet(s)).

6.4 Consideration of data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below should be developed and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph Number	Data Requirement Title
3.3.1	Propellant Analysis Report for Aircrew Escape Propulsion System (AEPS) Devices

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6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
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(Project 1377-0087)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-85097/1	**2. DOCUMENT DATE (YYMMDD)** 27 JUL 1998

3. DOCUMENT TITLE AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS) ROCKET CATAPULT MARK 16 MOD 1 (LOADED ASSEMBLY)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7. DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER, INDIAN HEAD DIVISION NAVAL SURFACE WARFARE CENTER (CODE 840M)	b. TELEPHONE *Include Area Code)* (1) Commercial 301-743-4700 (2) AUTOVON 354-4700
c. ADDRESS *(Include Zip Code)* 101 STRAUSS AVENUE. INDIAN HEAD, MD 20640-5035	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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